

# Comparison of established methods for quantifying genotyping error rates in wildlife forensics

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**Abstract** Several methods have been applied to calculate genotyping error rates (GER) for non-invasive population size estimations. However, there is a lack of comparability between these methods. Here we focused on the comparison of methods for determination of GER within one study using faeces samples of wild boars (*Sus scrofa*). Error rates were calculated by (1) comparison of reference tissue samples and rectum faeces samples (2) the number of deviations between replicates and the assumed consensus genotypes, (3) re-analysis of a subsample interpreted by allelic and genotype comparisons, and (4) a blind-test of anonymously subdivided faecal samples. The error rates differed widely between these four methods (0–57.5 %) and underline the need of a consensus approach. The blind-test resulted in a GER of 4.3 %. We recommend conducting such a blind-test for estimating realistic GER when starting a pilot study in wildlife forensics.

**Keywords** Non-invasive sampling · Population size estimation · Allelic dropout · False alleles · Multiple-tubes approach · Microsatellites

## Introduction

Population size estimation using non-invasive sampling has emerged as an important field in conservation genetics and is of crucial importance for wildlife management. Faeces as DNA source is attractive due to the potentially easy sampling and the homogeneous capture probability (Wehausen et al. 2004). However, faeces genotyping implicates pitfalls like reduced amplification success due to low target DNA concentrations, interference from diet and bacterial DNA, presence of PCR inhibitors (Lampa et al. 2008) or degradation of DNA (Idaghdour et al. 2003). Further challenges are the occurrence of incorrect genotypes due to allelic dropout or false alleles (Wehausen et al. 2004).

These genotyping errors can either lead to underestimations of population sizes when genetic tagging is not unique, and/or to overestimations when genotyping errors lead to additional false genotypes (Creel et al. 2003). Several studies were carried out to minimize genotyping errors, by e.g. using specific primers (Broquet et al. 2007), optimizing storage and extraction methods (Flagstad et al. 1999; Murphy et al. 2007), sampling fresh faeces (Maudet et al. 2004), thorough selection of genetic markers (Broquet et al. 2007), and applying a multi-tubes approach (MA; Taberlet et al. 1996).

However, genotyping errors can never be completely avoided in a microsatellite dataset, and therefore it is crucial to quantify the genotyping error rate (GER) in order to estimate the reliability of inferred results (Hoffman and Amos 2005; Pompanon et al. 2005). Valière et al. (2007) reviewed the commonly used genotyping error calculations,

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but without actually comparing the respective methods (see also Beja-Pereira et al. 2009; Broquet and Petit 2004). For this, a comparison of different methods for quantifying genotyping errors using the same data set is needed.

Here, we developed a general approach for achieving reliable microsatellite datasets for population size estimations in wildlife forensics with a realistic GER of less than 5 %, ensuring reliable population size estimations (Taberlet and Luikart 1999). It is important to consider the real GER instead of the theoretical GER because it reflects the accuracy of the individual assignment (Frantz et al. 2003). We genotyped 315 faecal samples from wild boars, calculated GER by applying different methods of MA and compared them. Finally, we present a blind-test for quantifying a realistic GER.

## Materials and methods

From 17 wild boars (*Sus scrofa*) obtained from a driving hunt in 2008 the Palatinate forest (SW Germany) tissue samples were taken from ear biopsies and faeces samples were taken from the rectum. Additionally, we collected 315 wild boar faecal samples in a 4000 ha area in the Palatinate forest (SW Germany) during a 3 week sampling period in December 2007. For the blind-test, eight additional faecal samples were collected within 1 day under the same conditions for method comparison. The tissue samples were stored in denatured ethanol 99.6 % at 4 °C and the rectum and faecal samples were stored in plastic bags and frozen at −20 °C until DNA extraction. Tissue samples were washed with 3 mL 1 × phosphate-buffered saline before DNA-isolation. The DNA was isolated using a Nucleo-Spin Tissue Kit (Macherey–Nagel), following the manufacturer's protocol. Approximately 250 mg of the surface of each faecal dropping directly used (without drying) for DNA-extraction, which was carried out according to the manufacturer's protocol from the Nucleo-Spin Tissue Kit (Macherey–Nagel) with the modification of three wash steps. We genotyped the DNA extracts from tissue and faeces with a subset of four species-specific microsatellite loci and additional sex marker (TNFB, Sw2496, Sw2021, Sw742 and PigSRY) which have proven good genotyping success in wild boar faeces samples (Kolodziej et al. 2011). The PCR was conducted in 15 µl containing (final concentration) 2.5 mM MgCl<sub>2</sub> for faeces, 1.5 mM MgCl<sub>2</sub> for tissue, 0.2 mM of each dNTP, 0.2 µM of each primer, 0.1 µg/µl bovine serum albumin (BSA only for faeces), 0.025 u/µl Go Taq Flexi DNA polymerase (Promega) and 1 µl template using a Primus 96 Cyclet (Peqlab Biotechnologie GmbH) under following conditions: initial denaturation at 95 °C for 2 min, followed by 40 cycles of 30 s at 95 °C, 30 s at locus specific annealing temperature and 30 s at 72 °C, and a terminal elongation step at 72 °C for 5 min. The amplification products were visualized on agarose gel. Samples with visible

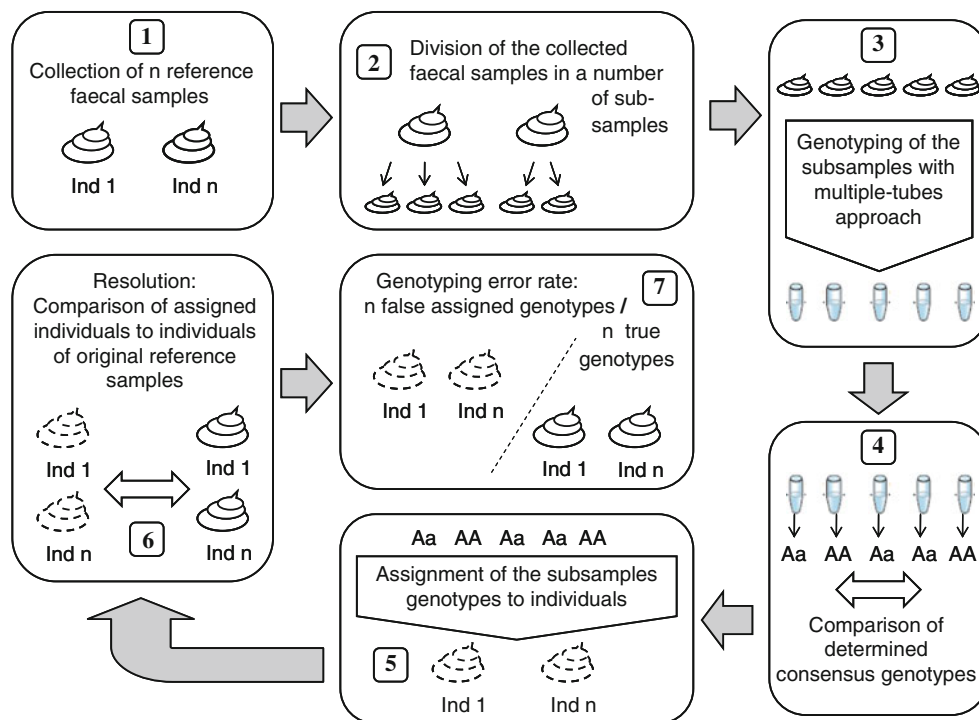
bands were run on a CEQ 8000 Sequencer (Beckman Coulter) and analyzed on the corresponding software CEQ SYSTEM 9.0 to determine allele lengths.

In a comparative MA (MA<sub>comp</sub>) PCR was repeated three to five times and individual alleles had to be identical across at least two repetitions for heterozygotes and three repetitions for homozygotes (Frantz et al. 2003). In case of five repeats with two heterozygotes and three homozygotes, the sample was assigned as heterozygote. In case of GER >5 % we altered the MA<sub>comp</sub> to a conservative MA (MA<sub>cons</sub>; Taberlet and Luikart 1999): homozygous genotypes had to be identical across at least eight repetitions. Samples with low genotyping success and ambiguous MA outcomes were discarded from the analysis. Consensus genotypes were determined using the software GIMLET (Valière 2002), with the default threshold value 2 for heterozygous, and 3 (MA<sub>comp</sub>) or 8 (MA<sub>cons</sub>) for homozygous genotypes. The consensus genotypes were manually scrutinized by re-checking the raw data of all one-mismatch pairs (Paetkau 2003).

Two sources of genotyping errors were defined: allelic dropout (ADO; one allele of a heterozygous locus was not amplified or not scored) and false allele (FA; an additional allele was misleadingly assigned to a heterozygous locus). Individual identification and genotype comparisons were calculated using the modified EXCEL sheet GENECAP (Wilberg and Dreher 2004). The following four methods were used to quantify GER: (1) Error rates were defined by using reference tissue samples and corresponding rectum faeces samples (n = 17). Tissue samples were analysed two times, whereas rectum faeces were analysed with MA<sub>comp</sub>. The determined tissue sample consensus genotypes were compared with the determined consensus genotypes of the rectum faeces. (2) The number of deviations from the consensus genotypes were counted for locus and individual by comparing the repeated genotypes and the assigned consensus genotype (Broquet and Petit 2004) using GIMLET; (3) A re-analysis (Hoffman and Amos 2005), starting from DNA-extraction, was conducted for a subsample of 10 % (N = 30) out of the whole sampling (N = 315). We compared the results by a) counting the allelic differences within consensus genotypes (Bonin et al. 2004) and b) comparing complete genotypes (Pompanon et al. 2005); (4) In a blind-test eight faecal samples from different wild boar individuals were anonymously divided into three to eight subsamples (N = 40) prior genotyping and subsequently assigned to an individual by matching the consensus genotype obtained from MA<sub>cons</sub>. The general procedure is demonstrated as a step by step scheme in Fig. 1.

## Results and Discussion

The major challenge in non-invasive genotyping is to minimize genotyping errors and to quantify the real GER.



**Fig. 1** Flowchart of a blind-test to estimate realistic genotyping error rates. The number of collected reference faecal samples should not be lower than 20. We recommend a minimum number of 10 % of collected samples in one collection period and to provide a higher number of samples from different individuals than subsamples. We

suggest starting the test with a comparative multiple-tubes approach (fewer repetitions) in respect of reduced costs. In case of high error rates, the multiple-tubes approach could be modified by increased repetitions

A reliable method for non-invasive real GER calculation is the use of reference samples like blood or tissue and corresponding faecal samples to obtain the true consensus genotypes (Bayes et al. 2000). Faeces samples should be collected directly after defecating and tissue samples taken of the same individual as reported in Bayes et al. (2000). In practice this procedure is often difficult to acquire and can only be suggested for species where hunting is permitted and hunted individuals are available. In our study reference faeces samples were directly removed from the rectum of shot boars. Comparisons of reference tissue samples and corresponding rectum faeces samples (method 1; Table 1) revealed no genotyping errors, thus supporting the MA<sub>comp</sub>.

Compared to method 1 we expected a higher GER in method 2, since in method 1 the final consensus genotypes were compared to each other whereas in method 2 the single genotypes obtained from the repeats were compared to an expected final consensus genotype of each sample. For faeces samples we assumed that environmental influences, like UV-light, endogenous endonuclease activity or oxidative damage, could be a reason for low quality of the samples (Deagle et al. 2006). Therefore we determined the GER with method 2. The MA<sub>comp</sub> used in method 2 resulted in 156 successful assignments out of 315 faeces samples, implying that 50 % of the samples were discarded

due to low DNA quality. This amount was in range with other non-invasive studies (Arrendal et al. 2007; Hedmark et al. 2004; Lucchini et al. 2002). The GER resulted in 26.8 % ADO and 7.3 % FA (Table 1). These values are in the range of those reported in comparable studies on otters (*Lutra lutra*; Hajkova et al. 2009; Ferrando et al. 2008), but lower than those for black rhinoceroses (*Diceros bicornis*; Garnier et al. 2001) and higher compared to a study on coyotes (*Canis latrans*; Prugh et al. 2005).

Our calculations for method 2 were initially based on deviations of the repeats to the consensus and not to the real genotype. Thus, we validated the results of method 2 with a re-analysis of 10 % of the samples (N = 30; method 3 using MA<sub>comp</sub>). Applying method 3a 216 of 240 allelic comparisons could be conducted. We identified 176 matches and 40 mismatches, which resulted in a considerably lower GER compared to method 2 (Table 1) and were in the same order of magnitude compared to a study on wolves (*Canis lupus*; Lucchini et al. 2002). According to method 3b, which is explained in the review of Pompanon et al. (2005), 23 of 30 consensus genotypes could be identified, and 10 out of 23 samples could be assigned to the correct individual, producing the highest GER in relation to the other methods (Table 1). To prove that the small sample size of N = 30 did not cause a bias, we calculated

**Table 1** Estimation of genotyping errors rates (GER) [%] based on allelic drop outs (ADO) and false alleles (FA) for comparative multiple-tubes approach (MA<sub>comp</sub>) and conservative multiple-tubes approach (MA<sub>cons</sub>) using three different methods

| Method | MA <sub>comp</sub> |     | MA <sub>cons</sub> |     |
|--------|--------------------|-----|--------------------|-----|
|        | ADO                | FA  | ADO                | FA  |
| 1      | 0                  | 0   | /                  | /   |
| 2      | 26.8               | 7.3 | 31.5               | 4.5 |
| 3a     | 16.5               | 2.0 | 0                  | 0.4 |
| 3b     | 49.0               | 8.0 | 0                  | 8.0 |
| 4      | /                  | /   | 0                  | 4.3 |

Method (1) Comparison of reference tissue samples and rectum faeces samples

(2) GER calculated by deviations of repeats to consensus genotypes

(3a) Re-analysis with GER calculated by allele comparisons

(3b) Re-analysis with GER calculated by genotype comparisons

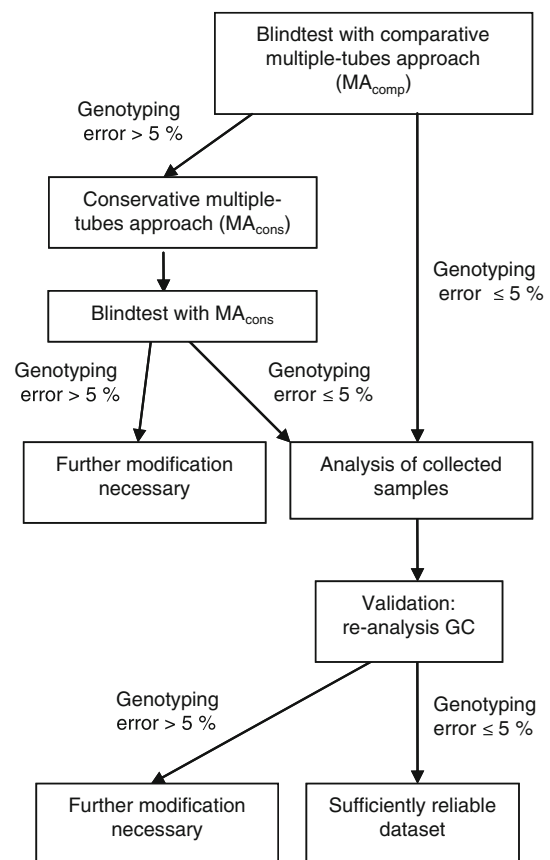
(4) Blind-test of anonymously subdivided faecal samples

the GER with method 2 for N = 30 (ADO = 22.7 %; FA = 7.3 %). These results were similar to the GER calculated with N = 315.

The discrepancies of GER estimations for our dataset (Table 1) were expected but astonishingly high. The high GER, mainly caused by ADO, showed that three identical repeats for a consensus genotype are not enough to reliably describe homozygous loci. ADO occurred considerably more often than previously assumed, and the MA<sub>cons</sub> method seems therefore preferable. However, after applying MA<sub>cons</sub>, method 3 provided an ADO rate of zero, whereas method 2 led to ADO rates higher than obtained from MA<sub>comp</sub> (Table 1). These results were due to the increased number of repetitions: a mis-identified consensus homozygous genotype (due to low number of repeats) obscures the actual ADO rate; in contrast this mis-identified homozygous genotype was correctly identified during a re-analysis and therefore leading to a more realistic ADO rate. This is an argument to resign from the calculation of GER by using the deviation of repeats from the assumed consensus genotype or by calculating the ratio of observed ADO using the number of heterozygous genotypes. These methods produce, however, only a theoretical GER.

Method 3b (genotype comparisons) is favourable compared to 3a (allele comparison) because a single error in a multilocus genotype already leads to a false classification of individuals and results in biased population size estimation (Harris et al. 2010). Therefore, the number of incorrect genotypes should be counted rather than simply counting incorrect alleles. To validate the results of method 3b, we conducted a blind-test (method 4) as control procedure, which is illustrated in Fig. 1. From eight collected samples (step 1) 40 faeces samples were divided (step 2) and 25 could be successfully genotyped (step 3 and 4). We

could identify six of eight different individuals (step 5). Resolution of the blind-test showed that 24 samples were correctly assigned to match original samples or individuals (step 6). Only one sample could not be assigned properly due to one FA. Thus, the blind-test resulted in the lowest GER of 4.3 % with ADO rate of 0 equal to results of method 3 (Table 1). This test implies a reliable and easily reproducible basis for calculating a realistic GER before starting a pilot study on population size estimates in wildlife forensics, especially when reference tissue or blood samples are not available. We recommend to use at least 20 faeces samples due to required statistical robustness (Bortz and Doering 2002) or if possible a minimum number of 10 % of collected samples in one collection period, which was suggested by Pompanon et al. (2005) for blind retyping, e.g. for re-analysis. It is important to provide a higher number of samples from different individuals than subsamples, due to the low DNA quality of faecal samples and thus a lower PCR success rate. This test can be



**Fig. 2** Flowchart presenting a general approach to obtain realistic genotyping error rates for reliable population size estimation in wildlife forensics. A genotyping error rate of 5 % is assumed to be adequate for population size estimation. The blind-test can be used in a pilot study to test multiple-tubes approaches and can be validated with re-analysis GC (genotype comparison) after analysis of collected faecal samples for a study

transferred to other systems using different non-invasive samples (hairs or feather) with the assumption that samples are dividable or the sampling provides more than one sample per individual.

In conclusion, calculated GERs are often incomparable between studies due to the different approaches used. Therefore, different expected outcomes even for using the same approach but a different calculation are possible (see difference between genotype and allele comparison). Here, we want to highlight the importance to scrutinize the reliability of genotyping results provided by using only one method. Furthermore, we proved the necessity to calculate a realistic GER, opposed to a theoretical GER, which is based on the deviation of acquired genotyping repeats to an expected consensus genotype. This is important for studies dealing with e.g. population size estimations or mating system analyses. Finally, in Fig. 2 we recommend a general stepwise approach for obtaining a realistic GER for reliable population size estimation in wildlife forensics.

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